

MOSAIC: single-cell expression from H&E histology

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Abstract

Spatial transcriptomics maps gene expression in tissue, but cost and throughput limit large studies of ulcerative colitis and related disorders. Here we present MOSAIC, a two-phase computational framework that predicts single-cell gene expression from routine H&E histology and then assigns cell lineages using only those predicted profiles. Phase A uses each cell's image together with six spatially nearest neighbours; Phase B classifies nine curator-defined in-house lineages (integrative clustering of challenge expression; not supplied as challenge annotations) without further use of images. On inflamed colon tissue from the Autoimmune Disease Machine Learning Challenge, with training and test on different patient sections, mean agreement between predicted and measured expression was modest (correlation 0.21 across 460 genes) with heterogeneous recovery across genes, and lineage accuracy reached 60%, well above chance but below a control trained on measured expression (95%). These findings show partial recovery of molecular signal from histology under conservative evaluation, not clinical utility or generalisation beyond this dataset.

Keywords: spatial transcriptomics, histopathology, deep learning, ulcerative colitis, attention

1 Introduction

Spatial transcriptomics measures gene expression while preserving where cells sit in tissue [1, 2]. The approach has advanced understanding of tissue organisation and disease, yet cost and technical demands still limit how widely it can be applied. In ulcerative colitis (UC), pathologists rely on H&E-stained biopsies to assess inflammation and healing, whereas single-cell and spatial genomics reveal immune, epithelial and stromal programmes that morphology alone does not fully resolve [3–5]. Archival H&E collections are vast compared with paired spatial assays, motivating models that infer cell-resolved expression from routine histology.

Computational methods now link histology to spatial or single-cell expression [6]. Spot-level models aggregate many cells per measurement [7, 8]; single-cell approaches use contrastive learning, multiple-instance designs or multitask heads [9, 10]. Neighbourhood size, aggregation and validation design differ across studies, and reported accuracy often depends on random cell-level splits that can inflate scores when cells from the same slide appear in both training and test—a form of spatial leakage that overstates generalisation to new sections and patients.

Here we present MOSAIC (Morphology-Oriented Spatial Attention for Inferring Cell Expression), a two-phase framework evaluated on challenge colon tissue with Xenium readouts. **Phase A** predicts a 460-gene panel from the target cell’s H&E patch and $K=6$ spatial nearest neighbours, using a convolutional image encoder and attention to weight neighbours. **Phase B** assigns nine integration-derived lineages from predicted expression only, without images at test time, to test whether Phase A retains lineage-relevant molecular signal beyond aggregate correlation. We use one patient-disjoint split (all UC6 for training; 20% of UC7 for validation; 10% of UC1 for testing; seed 42) so that no cells from validation or test slides enter Phase A training. Detailed outcomes and limitations are reported in Results and Methods.

2 Results

We report one pre-specified patient-disjoint evaluation on challenge colon data: histology-to-expression recovery (Phase A) and lineage assignment from predicted profiles alone (Phase B).

2.1 Study design and data

Data come from the Autoimmune Disease Machine Learning Challenge (Broad Institute, 2024)¹, with co-registered H&E and Xenium in situ spatial transcriptomics [11] from human colon. Each cell has a challenge panel of **460 genes**, a 128×128 pixel H&E patch centred on the nucleus, a 460-dimensional expression vector and, for Phase B,

¹Official name: Autoimmune Disease Machine Learning Challenge (AMLC).

an integration-derived lineage label. Inflamed ulcerative colitis sections UC6, UC7 and UC1 define the primary split.

Expression was $\log(1 + x)$ transformed and z-scored using means and standard deviations fit on **training cells (UC6)** only. Cell-type labels were produced in house by integrating all seven challenge samples, Leiden clustering [12] in Scanpy [13], and curator mapping to nine lineages: B cells, endothelial cells, epithelial cells, fibroblasts, glia, lymphatic endothelial cells, myeloid cells, T cells and vascular smooth muscle. These labels are operational supervision targets, not independent histopathology for every cell.

The primary split enforces slide-level separation: Phase A trains on **all** UC6 cells, monitors on a random **20%** of UC7 and tests on a random **10%** of UC1 (seed 42). After requiring paired patches and labels, $n=223,781$ (train), 28,941 (validation) and 20,253 (test) cells remain (Extended Data Table 1). Phase B uses the same partition on Phase A outputs. Patches are in pixel coordinates; microns-per-pixel was not provided in the challenge release.

We prioritise slide-disjoint generalisation over pooled cell-level K -fold cross-validation, which can inflate scores when correlated cells from the same section appear in both training and test. Uncertainty from the single test subsample is not quantified by resampling or bootstrap in this study.

2.2 Gene expression from histology (Phase A)

Phase A predicts the 460-gene panel from H&E and $K=6$ spatial neighbours with attention-based aggregation (Methods). On UC1 test cells, mean per-gene Pearson correlation (PCC) between predicted and measured Xenium expression was **0.205** (median 0.184; Extended Data Table 2). Under slide-disjoint hold-out this indicates partial, not complete, recovery of the panel from morphology. Gene-level PCC was heterogeneous: several structural and epithelial markers, including *TAGLN* and *EPCAM*, ranked among the highest per-gene correlations (Extended Data Table 9), consistent with morphology-linked recovery for subsets of the panel rather than uniform reconstruction. Magnitudes are lower than many literature values obtained under random cell-level splits (Extended Data Table 8), underscoring that conservative evaluation materially reduces apparent accuracy relative to pooled cell-level protocols. Validation monitoring on UC7 reached global correlation 0.263 over flattened gene-cell pairs; that value is not used for test claims.

2.3 Lineage assignment from predicted expression (Phase B)

Phase B is a molecular utility test: it assigns nine lineages from **predicted** expression only—no H&E at test time—to ask whether Phase A outputs retain lineage-relevant structure. On UC1 test cells ($n=20,253$), uniform chance and majority-class baselines yield 11.1% and 14.4% accuracy. MOSAIC reaches **59.9%** accuracy (macro F1 0.514), well above chance. The same multi-layer perceptron trained on *measured* Xenium expression under the identical split reaches **94.7%** accuracy, a ceiling control showing that labels are separable in the panel when histology-induced error is removed and that expression reconstruction currently limits downstream typing.

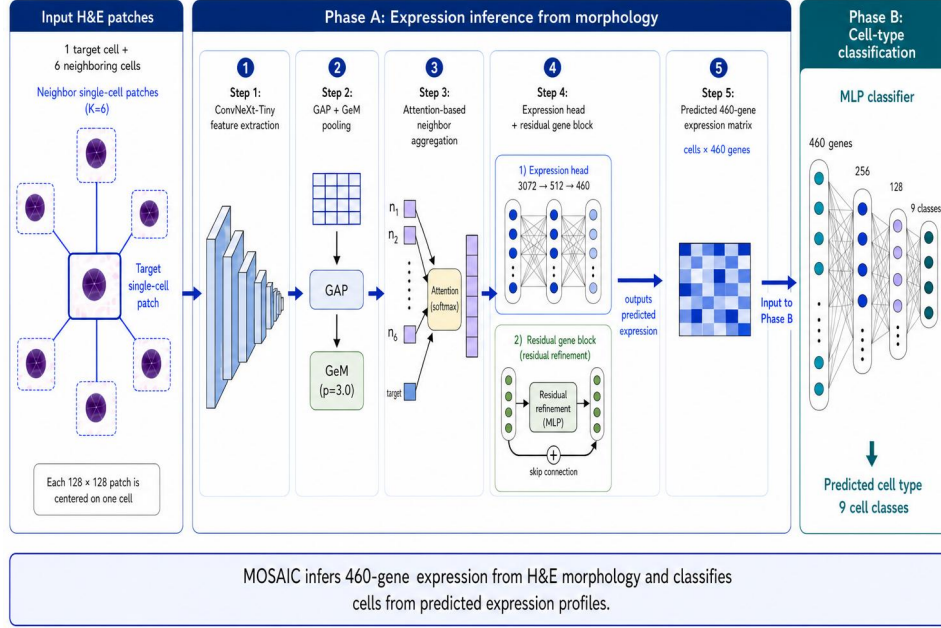


Fig. 1 Overview of the MOSAIC two-phase framework. **a**, Phase A: 128×128 px patches from a target cell and $K=6$ spatial neighbours are encoded by ConvNeXt-Tiny; attention aggregates neighbour embeddings; an expression head outputs 460 genes. **b**, Phase B: an MLP ($460 \rightarrow 256 \rightarrow 128 \rightarrow 9$) classifies nine lineages from predicted expression only (no H&E at test time).

Table 1 Phase B classification on UC1 test cells ($n=20,253$). Same patient-disjoint split and MLP architecture (Methods); rows differ only in input expression.

Input to MLP	Acc.	Macro F1
Random (uniform over nine lineages)	0.111	0.111
Majority class (always B cells)	0.144	0.028
MOSAIC Phase B: predicted expression	0.599	0.514
Measured Xenium (ceiling control) [†]	0.947	0.945

[†] Observed 460-gene matrix from in situ profiling, not inferred from H&E.

111 Per-class metrics are uneven (Extended Data Table 3); rare lymphatic endothelial
 112 cells show low recall. Section-shifted class frequencies (Extended Data Table 4) affect
 113 weighted versus macro F1.

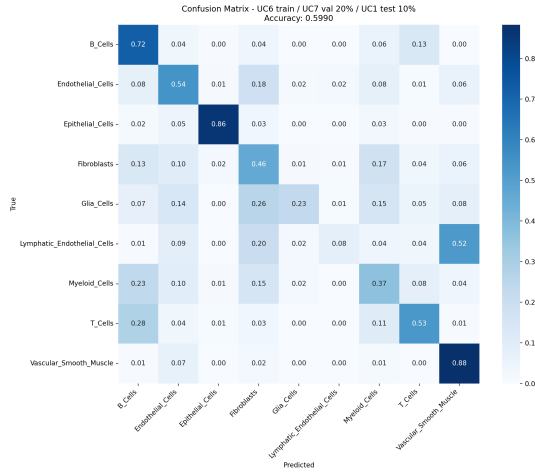


Fig. 2 Phase B confusion matrix from predicted expression (UC1 test cells). Row-normalised proportions (rows: true label; columns: predicted). Primary MOSAIC endpoint; same split as above.

2.4 Discussion

On one patient-disjoint colon split with $K=6$ neighbours, MOSAIC delivers partial recovery of a 460-gene panel from H&E (mean PCC 0.205) and moderate lineage assignment from predicted expression alone (59.9% accuracy). Under slide-disjoint hold-out these outcomes support a UC-focused proof-of-concept that neighbourhood context and two-phase supervision can link routine histology to single-cell molecular readouts on inflammatory colon tissue. They are not headline clinical benchmarks and do not establish utility on unseen cohorts.

Three layers support interpreting this proof-of-concept. First, evaluation prioritises transfer across patients and sections rather than pooled cell-level cross-validation, reducing spatial leakage between training and test. Second, Phase B tests molecular utility of Phase A: lineages must be inferred from predicted profiles without image features. Third, the pipeline is fully specified—normalisation fit on UC6 training cells only, early stopping on UC7, test metrics on UC1 only, and deterministic training (seed 42) with exact counts in Extended Data—which facilitates replication on the same challenge release. Gene-level correlation was heterogeneous (for example, *TAGLN* and *EPCAM*); we do not treat top genes as independent biological validation.

Discordance between the measured-expression ceiling (94.7%) and the primary Phase B endpoint (59.9%) indicates that Phase A fidelity limits typing more than label definition in measured space. Limitations include a single train/validation/test configuration without bootstrap confidence intervals; no systematic ablations over neighbourhood size or backbone; dependence on challenge preprocessing, panel coverage, H&E-Xenium alignment and in-house labels; and undocumented microns-per-pixel. Multi-section benchmarks, uncertainty quantification and orthogonal readouts will be needed to assess translational relevance in UC and related disease.

3 Methods

3.1 MOSAIC overview

Neighbourhood construction. For each cell, we identify $K=6$ nearest neighbours by Euclidean distance in section coordinates (configurable K ; all experiments use $K=6$). Extended Data Figure 1 illustrates target centroids and neighbour sets. Each cell contributes one 128×128 RGB H&E patch centred on the nucleus for itself and for each neighbour [14].

Phase A encoder and attention. Patches are encoded with ConvNeXt-Tiny [15] initialised from ImageNet [16]. Global average pooling and generalised mean pooling (learnable p , initialised to 3.0) [17] are concatenated to 1536 dimensions per image. The target embedding defines a query; neighbour embeddings define keys and values. Scaled dot-product attention [18, 19] yields weights and context:

$$\alpha_j = \frac{\exp(\mathbf{Q} \cdot \mathbf{K}_j^\top / \sqrt{d})}{\sum_{k=1}^K \exp(\mathbf{Q} \cdot \mathbf{K}_k^\top / \sqrt{d})}, \quad \mathbf{c} = \sum_{j=1}^K \alpha_j \mathbf{V}_j \quad (1)$$

where $d = 1536$ is the projected feature dimension. For each neighbour j , the score $\mathbf{Q}\mathbf{K}_j^\top$ measures compatibility between the target query and that neighbour’s key; the softmax produces normalised weights α_j with $\sum_j \alpha_j = 1$, so $\mathbf{c}$ is a convex combination of neighbour value vectors—learned mixing on the K NN spatial graph rather than a fixed average. Attention is restricted to the pre-selected K nearest neighbours. Target and context are concatenated (3072 dimensions), passed through a linear–ReLU–dropout expression head (460 genes) and a residual gene block (two linear–GELU–linear sub-blocks with a skip connection). Full layer dimensions and hyperparameters are in Extended Data Table 5.

Phase B classifier. A multi-layer perceptron maps 460-dimensional Phase A outputs to nine lineage logits ($460 \rightarrow 256 \rightarrow 128 \rightarrow 9$; ReLU and dropout 0.5). Inputs are z-scored using Phase A prediction statistics on UC6 training cells. No image features are used at inference. Extended Data Table 7 summarises qualitative differences from related histology-to-expression methods.

3.2 Training and evaluation

Split. Phase A trains on all UC6 cells; early stopping uses mean per-gene PCC on a 20% UC7 subsample (UC7 labels are not used for test-set claims or hyperparameter tuning on UC1); test metrics use a 10% UC1 subsample (seed 42). No cells from UC7 or UC1 enter Phase A training, avoiding split leakage from random cell-level mixing within slides. Phase B trains on UC6 Phase A outputs, validates on the UC7 subsample (macro F1) with the same early-stopping rule, and tests on the UC1 subsample with frozen Phase A predictions. A diagnostic control retrain the same MLP on observed Xenium vectors.

Phase A optimisation. Mean squared error on log-normalised, z-scored expression; AdamW (learning rate 2×10^{-4} , weight decay 10^{-4}); batch size 64; up to 30

epochs; early stopping after six epochs without validation PCC improvement; learning-rate reduction on plateau (factor 0.3, patience 2); exponential moving average of weights (decay 0.9995). Augmentation parameters are in Extended Data Table 6; ImageNet normalisation is applied to patches.

Phase B optimisation. Cross-entropy; Adam (10^{-3} , weight decay 10^{-4}); batch size 128; up to 30 epochs; early stopping after five epochs without validation macro F1 improvement.

Implementation. Deterministic training (seed 42) in PyTorch [20]. We do not report multi-seed replication, bootstrap intervals or systematic ablations over K , aggregation mode or backbone.

3.3 Statistics

Phase A. For gene g on a specified cell set, PCC is the Pearson correlation between predicted and observed expression across cells. The primary endpoint is the mean of gene-wise PCC over 460 genes on UC1 test cells. Global correlation on UC7 validation cells (flattened gene-cell pairs) is used only for training monitoring.

Phase B. Accuracy, macro F1 (unweighted mean of per-class F1), weighted F1 and balanced accuracy on UC1 test cells. No hypothesis tests, resampling or bootstrap were performed; cell counts n are reported exactly for the single split described above.

3.4 Expression preprocessing and cell-type annotation

The Autoimmune Disease Machine Learning Challenge (AMLC) release provides co-registered H&E whole-slide mosaics, Xenium in situ spatial transcriptomics, a fixed **460-gene** panel, per-cell expression matrices and 128×128 pixel H&E patches centred on nuclei. It does *not* provide finished single-cell type labels for supervised classification.

We produced lineage labels in house. Using all seven AMLC colon samples (inflamed, I, and non-inflamed, NI), we integrated and harmonised transcriptomic profiles across specimens to attenuate inter-sample technical variation and batch effects. Unsupervised clustering used the Leiden algorithm [12] in Scanpy [13], followed by cluster-level differential expression. Nine functional lineages were assigned by manual curation, mapping clusters to B cells, endothelial cells, epithelial cells, fibroblasts, glia, lymphatic endothelial cells, myeloid cells, T cells and vascular smooth muscle using canonical marker genes. For model training, expression was $\log(1+x)$ transformed and z-scored with means and standard deviations fit on UC6 training cells only (Results).

These labels are *operational* supervision targets for Phase B. They reflect integrative transcriptomic clustering and curator decisions, not an independent histopathology reference for every cell, and inherit uncertainty from integration, resolution and ambiguous markers.

3.5 Data availability

The H&E images and matched Xenium spatial transcriptomics data used in this study were obtained from the Autoimmune Disease Machine Learning Challenge (AMLC), an openly documented computational challenge organised by the Broad Institute of

MIT and Harvard and hosted on the CrunchDAO platform (<https://docs.crunchdao.com/competitions/competitions/broad-institute-autoimmune-disease>). The dataset is registration-gated: it is openly documented but distributed only to registered participants under the applicable challenge and platform terms. The present study used these data solely for secondary computational analysis and did not redistribute the raw challenge files (H&E images, Xenium files, image patches or expression matrices). Researchers interested in accessing the data should consult the official challenge page and the applicable CrunchDAO and challenge terms. The same dataset has been used in a recent methodological benchmark [14].

3.6 Code availability

Code for preprocessing, training and evaluation will be deposited in a public repository upon publication and is available to reviewers on request.

3.7 Large language models

Large language models (e.g. ChatGPT) were used solely to assist with language editing and revision of the manuscript text. They were not used for data analysis, model training, result generation or scientific interpretation.

Author contributions. T.F.: software, formal analysis, writing—review and editing; C.C.: methodology, supervision; H.T.I.N.: conceptualization, supervision; A.S.: resources; M.C.C.M.: resources, supervision; M.F.V.: investigation; G.R.M.: formal analysis.

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Ethics statement. This study involved only secondary computational analysis of data obtained from the Autoimmune Disease Machine Learning Challenge, an openly documented computational challenge organised by the Broad Institute of MIT and Harvard and hosted on the CrunchDAO platform. The data were distributed to registered participants under the applicable challenge and platform terms and did not contain identifiable patient information. The authors did not recruit human participants, collect human tissue samples, generate new human-subject data or access identifiable patient information; the original tissue collection was performed by the data-generating institution under the ethical oversight applicable to that study. Accordingly, no additional institutional ethics approval or informed consent was required for the analyses reported in this manuscript. Cell-type labels used for supervised classification were derived in house from challenge expression data and were not provided as annotations by the challenge organisers. No new human participants or biological specimens were recruited for this study.

Field	Value
train_image / val_image / test_image	UC6 / UC7 / UC1
val_fraction / test_fraction	0.2 / 0.1
train_size / val_size / test_size	223781 / 28941 / 20253
expr_val_pearson	0.2629683796571803
expr_test_pearson	0.20515923420662494
cls_val_f1	0.5910889339765822
test_accuracy	0.5990223670567323
test_f1_macro	0.5143238982672779
test_f1_weighted	0.5928601117645256

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Appendix A Extended Data

Extended Data figures and tables are published online only and are not included in the print/PDF version of the Article.



Fig. A1 Extended Data Fig. 1 — Spatial neighbourhood selection. **(a)** Target cell centroid. **(b)** $K=6$ nearest neighbours in section coordinates.

A.1 Extended Data Table 1

Primary experimental run (exact values).

This table reports the exact metrics exported by the primary training run described in the main text ($K=6$, attention aggregation, section-level split with train/validation/test on UC6, UC7 and UC1; challenge image identifiers UC6.I, UC7.I, UC1.I).

269 A.2 Extended Data Table 2

270 Gene-level PCC distribution on UC1 test split.

271 To complement the mean PCC reported in the main text, we computed gene-level
 272 Pearson correlation coefficients for all 460 genes on the UC1.I test subset ($n = 20,253$
 273 cells; 10% of cells from that section under the fixed split). The distribution had mean
 274 PCC 0.2052, median PCC 0.1845 and standard deviation 0.1493.

Statistic	Value
Number of test cells	20,253
Number of genes	460
Mean PCC across genes	0.2052
Median PCC across genes	0.1845
Standard deviation of gene-level PCC	0.1493

275 A.3 Extended Data Table 3

276 Per-class classification metrics (UC1 test).

277 Per-class metrics for the primary Phase B endpoint (MOSAIC on predicted expres-
 278 sion; UC1 test cells, $n=20,253$). Balanced accuracy reported in the main text (0.520)
 279 equals the macro-averaged recall.

Cell type	Precision	Recall	F1	Support
B cells	0.481	0.721	0.577	2915
Endothelial cells	0.379	0.543	0.447	1517
Epithelial cells	0.859	0.865	0.862	1138
Fibroblasts	0.458	0.459	0.459	2659
Glia cells	0.427	0.229	0.298	497
Lymphatic endothelial cells	0.153	0.081	0.106	161
Myeloid cells	0.583	0.367	0.451	4523
T cells	0.578	0.529	0.553	2342
Vascular smooth muscle	0.872	0.883	0.877	4501
Macro avg	0.532	0.520	0.514	20,253
Weighted avg	0.609	0.599	0.593	20,253

280 A.4 Extended Data Table 4

281 Cell counts by type (UC6 train vs. UC1 test).

282 Counts after filtering to cells with paired patches and in-house lineage labels
 283 under the fixed split (seed 42). The challenge release did not include finished cell-type
 284 annotations; labels follow the integrative curation pipeline in Methods (Section 3.4).

Cell type	n UC6 train	n UC1 test
Myeloid cells	21,252	4523
Vascular smooth muscle	23,822	4501
B cells	42,324	2915
Fibroblasts	30,082	2659
T cells	39,378	2342
Endothelial cells	24,823	1517
Epithelial cells	34,884	1138
Glia cells	5330	497
Lymphatic endothelial cells	1886	161
Total	223,781	20,253

285 A.5 Extended Data Table 5

Table A1 Extended Data Table 5 — Architecture details and hyperparameters of MOSAIC Phase A.

Component	Configuration	Output Dimension
<i>Visual Feature Extraction</i>		
Backbone	ConvNeXt-Tiny (ImageNet pretrained)	768
Global Average Pooling	Spatial mean	768
Generalised Mean Pooling	Learnable p (init=3.0)	768
Multi-scale Features	GAP GeM concatenation	1536
<i>Attention Mechanism</i>		
Query Projection	Linear(W_Q)	1536
Key Projection	Linear(W_K)	1536
Value Projection	Linear(W_V)	1536
Attention Type	Scaled dot-product	—
<i>Expression Prediction</i>		
Input	Target features Context	3072
Hidden Layer	Linear + ReLU + Dropout(0.1)	512
Output Layer	Linear	460
Gene Block	$2 \times (\text{Linear} + \text{GELU} + \text{Linear}) + \text{Residual}$	460

Table A2 Extended Data Table 6 — Data augmentation parameters for training.

Transformation	Parameter	Value
Horizontal Flip	Probability	0.5
Vertical Flip	Probability	0.5
Random Rotation	Max angle	$\pm 5^\circ$
Color Jitter	Brightness	± 0.03
Color Jitter	Contrast	± 0.03
Color Jitter	Saturation	± 0.03
Color Jitter	Hue	± 0.01

Table A3 Extended Data Table 7 — Comparison of MOSAIC with related methods for spatial gene expression prediction.

Method	Resolution	H&E	Spatial Neigh.	Pos. Enc.	Attention	K	Backbone
ST-Net	Spot	✓	×	×	×	—	DenseNet
HisToGene	Spot	✓	×	Learned	✓	—	ViT
His2ST	Spot	✓	✓	Distance	✓	4	CNN+Spatial
THItToGene	Spot	✓	✓	Distance	✓	4	DynConv+Attn
HGGEP	Spot	✓	✓	Dist+Pos	×	K	Hyper
BLEEP	Spot	✓	×	×	×	—	Contrastive
DANet	Spot	✓	×	×	×	—	SSM
PixNet	Dense	✓	×	×	×	—	UNet
HEIST	Cell	×	✓	Sinusoidal	✓	K	Transformer
GHIST	Cell	✓	×	×	✓	—	Multitask
HistoCell	Cell	✓	✓	×	×	—	Weakly-sup
SciSt	Cell	✓	×	×	✓	—	ResNet+SAM
sCellST	Cell	✓	×	×	×	—	MIL
ROSIE [†]	Cell	✓	×	×	×	—	ConvNeXt
MOSAIC	Cell	✓	✓	×	✓	6	ConvNeXt

[†]ROSIE predicts protein expression (50 biomarkers) rather than gene expression.

286 A.6 Extended Data Table 6

287 A.7 Extended Data Table 7

288 A.8 Extended Data Table 8

289 A.9 Extended Data Table 9

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Table A4 Extended Data Table 8 — Illustrative literature context for histology-to-expression PCC (settings differ; not a head-to-head benchmark).

Method	PCC (context)	# Genes
ST-Net [7]	$\sim 0.15\text{--}0.25^*$	785
HisToGene [21]	$\sim 0.15\text{--}0.25^*$	785
His2ST [8]	$\sim 0.20\text{--}0.26^*$	785
THltoGene [22]	$\sim 0.20\text{--}0.25^*$	785
EGNv2 [†] [6]	0.28	785
DeepPT [†] [6]	$\sim 0.25\text{--}0.31^*$	785
BLEEP [23]	$\sim 0.25^*$	3467
GHIST [10]	0.16 (spot); $\sim 0.6\text{--}0.7$ (SVGs)	280–377
sCellST [9]	~ 0.20 (HVGs)	50–500
MOSAIC (UC6→UC1 test)	0.205	460

*Approximate range from benchmark [6] (HER2+ ST dataset). [†]EGNv2 and DeepPT values reproduced from the benchmark compilation. MOSAIC: mean per-gene PCC on UC1 test subset (Section 2.1).

Table A5 Extended Data Table 9 — Ten genes with highest per-gene PCC on UC1 test cells.

Rank	Gene	PCC
1	TAGLN	0.8272
2	MYLK	0.8182
3	DES	0.7998
4	ACTA2	0.7323
5	TPM2	0.7020
6	EPCAM	0.6925
7	KRT19	0.6119
8	DMBT1	0.6087
9	DUOX2	0.5692
10	PLN	0.5690

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